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1 Connics

1.1 Conic Sections and Conics

Problem 1

Determine the equation of each of the circles with the following centre and radius.

Solution. (a) Centre the origin, radius 1:

$$x^2 + y^2 = 1.$$

(b) Centre the origin, radius 4:

$$x^2 + y^2 = 16.$$

(c) Centre (3, 1), radius 2:

$$(x - 3)^2 + (y - 1)^2 = 4.$$

(d) Centre (3, 4), radius 3:

$$(x - 3)^2 + (y - 4)^2 = 9.$$

Problem 2

Determine the condition on the numbers f , g , and h in the equation

$$x^2 + y^2 + fx + gy + h = 0$$

for the circle to pass through the origin.

Solution. Clearly $(0, 0)$ must be a solution, so we require

$$0^2 + 0^2 + f(0) + g(0) + h = 0,$$

which gives $h = 0$. Thus the condition is $h = 0$.

Problem 3

Determine the centre and radius of the circles given by the following equations.

Solution. (a) $x^2 + y^2 - 2x - 6y + 1 = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-2), -\frac{1}{2}(-6)\right) = (1, 3)$$

and radius

$$\sqrt{\frac{1}{4}(-2)^2 + \frac{1}{4}(-6)^2 - 1} = \sqrt{1 + 9 - 1} = 3.$$

(b) $3x^2 + 3y^2 - 12x - 48y = 0$. Dividing through by 3 gives $x^2 + y^2 - 4x - 16y = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-4), -\frac{1}{2}(-16)\right) = (2, 8)$$

and radius

$$\sqrt{\frac{1}{4}(-4)^2 + \frac{1}{4}(-16)^2} = \sqrt{4 + 64} = \sqrt{68} = 2\sqrt{17}.$$

Problem 4

Determine the set of points (x, y) in $\mathbb{R}^2$ that satisfies each of the following equations.

Solution. (a) $x^2 + y^2 + x + y = 1$. By Theorem 2 we have centre $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ and radius

$$\sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}.$$

So the set is all points $(x, y) \in \mathbb{R}^2$ with distance $\sqrt{\frac{3}{2}}$ from $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, i.e. a circle.

(b) $x^2 + y^2 - 2x + 4y + 5 = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-2), -\frac{1}{2}(4)\right) = (1, -2)$$

and radius

$$\sqrt{\frac{1}{4}(-2)^2 + \frac{1}{4}(4)^2 - 5} = \sqrt{1 + 4 - 5} = 0.$$

So the set is the single point $(1, -2)$.

(c) $2x^2 + 2y^2 + x - 3y - 5 = 0$. Dividing through by 2 gives

$$x^2 + y^2 + \frac{1}{2}x - \frac{3}{2}y - \frac{5}{2} = 0.$$

By Theorem 2 we have centre

$$\left(-\frac{1}{2}\left(\frac{1}{2}\right), -\frac{1}{2}\left(-\frac{3}{2}\right)\right) = \left(-\frac{1}{4}, \frac{3}{4}\right)$$

with radius

$$\sqrt{\frac{1}{4}\left(\frac{1}{2}\right)^2 + \frac{1}{4}\left(\frac{3}{2}\right)^2 + \frac{5}{2}} = \sqrt{\frac{1}{16} + \frac{9}{16} + \frac{40}{16}} = \sqrt{\frac{50}{16}} = \frac{5\sqrt{2}}{4}.$$

So the set is all points $(x, y) \in \mathbb{R}^2$ with distance $\frac{5\sqrt{2}}{4}$ from $\left(-\frac{1}{4}, \frac{3}{4}\right)$, i.e. a circle.

Problem 5

Determine which, if any, of the following pairs of intersecting circles are mutually orthogonal.

Solution. (a) $C_1 = \{(x, y) \mid x^2 + y^2 - 4x - 4y + 7 = 0\}$ and $C_2 = \{(x, y) \mid x^2 + y^2 + 2x - 8y + 5 = 0\}$. Notice

$$(-4)(2) + (-4)(-8) = -8 + 32 = 24, \quad 2(7 + 5) = 2(12) = 24.$$

So by Theorem 3, C_1 and C_2 are orthogonal.

(b) $C_1 = \{(x, y) \mid x^2 + y^2 + 3x - 6y + 5 = 0\}$ and $C_2 = \{(x, y) \mid 3x^2 + 3y^2 + 4x + y - 15 = 0\}$. Dividing through by 3 on C_2 gives

$$C_2 = \{(x, y) \mid x^2 + y^2 + \frac{4}{3}x + \frac{1}{3}y - 5\}.$$

Notice

$$(3)(\frac{4}{3}) + (-6)(\frac{1}{3}) = 4 - 2 = 2, \quad 2(5 + (-5)) = 2(0) = 0.$$

Since $2 \neq 0$, by Theorem 3, C_1 and C_2 are not orthogonal. ■

Problem 6

Find the equation of the line through the points of intersection of the circles

$$x^2 + y^2 - 3x + 4y - 1 = 0, \quad x^2 + y^2 + 5x - 6y + 3 = 0.$$

Solution. By Theorem 4 the equation of the line is the difference of the two circle equations:

$$(x^2 + y^2 - 3x + 4y - 1) - (x^2 + y^2 + 5x - 6y + 3) = 0,$$

which simplifies to

$$-8x + 10y - 4 = 0, \quad \text{i.e.} \quad 4x - 5y + 2 = 0. \quad \text{■}$$

Problem 7

This question concerns the parabola E with equation $y^2 = x$ and parametric equations $x = t^2, y = t$ where $t \in \mathbb{R}$.

(a) Write down the focus, vertex, axis, and directrix of E .

Solution. Writing $y^2 = 4ax = x$, so $4a = 1 \implies a = \frac{1}{4}$. So it has focus $(\frac{1}{4}, 0)$ and directrix $x = -\frac{1}{4}$. Its vertex is $(0, 0)$ and its axis is the x -axis.

Let P and Q be coordinates on E so that

$$P = (t_1^2, t_1), \quad Q = (t_2^2, t_2).$$

If $t_1 \neq t_2$ then the chord PQ has slope

$$\frac{t_2 - t_1}{t_2^2 - t_1^2} = \frac{1}{t_2 + t_1}.$$

Since (t_1^2, t_1) lies on the line we have

$$(y - t_1) = \left(\frac{t_2 - t_1}{t_2^2 - t_1^2} \right) (x - t_1^2) \implies (y - t_1)(t_2^2 - t_1^2) = (t_2 - t_1)(x - t_1^2) \implies (y - t_1)(t_2 + t_1) = (x - t_1^2).$$

It passes through the focus $(\frac{1}{4}, 0)$ iff

$$(0 - t_1)(t_2 + t_1) = \left(\frac{1}{4} - t_1^2 \right) \implies -t_1 t_2 - t_1^2 = \frac{1}{4} - t_1^2 \implies -t_1 t_2 = \frac{1}{4} \implies t_1 t_2 = -\frac{1}{4}.$$

So the chord PQ is a focal chord precisely when $t_1 t_2 = -\frac{1}{4}$. ■

Problem 8

Let P be an arbitrary point on the ellipse with equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

and focus $F(ae, 0)$. Let M be the midpoint of FP . Prove that M lies on an ellipse whose center is midway between the origin and F .

Solution. We now prove M lies on the ellipse

$$\frac{\left(x - \frac{ae}{2}\right)^2}{\left(\frac{a}{2}\right)^2} + \frac{y^2}{\left(\frac{b}{2}\right)^2} = 1,$$

which is centered at $\left(\frac{ae}{2}, 0\right)$, the midpoint of the origin and F . Parametrize the original ellipse by $P = (a \cos t, b \sin t)$ with $t \in \mathbb{R}$, so every point P on it has this form. With $F = (ae, 0)$, the midpoint of FP is

$$M = \left(\frac{ae + a \cos t}{2}, \frac{b \sin t}{2} \right).$$

Substituting the coordinates of M into the left-hand side, we have

$$x - \frac{ae}{2} = \frac{ae + a \cos t}{2} - \frac{ae}{2} = \frac{a \cos t}{2},$$

so

$$\frac{\left(x - \frac{ae}{2}\right)^2}{\left(\frac{a}{2}\right)^2} = \frac{\left(\frac{a \cos t}{2}\right)^2}{\left(\frac{a}{2}\right)^2} = \cos^2 t, \quad \frac{y^2}{\left(\frac{b}{2}\right)^2} = \frac{\left(\frac{b \sin t}{2}\right)^2}{\left(\frac{b}{2}\right)^2} = \sin^2 t.$$

Adding these gives

$$\frac{\left(x - \frac{ae}{2}\right)^2}{\left(\frac{a}{2}\right)^2} + \frac{y^2}{\left(\frac{b}{2}\right)^2} = \cos^2 t + \sin^2 t = 1.$$

■

Problem 9

Let P be a point $(\sec t, \frac{1}{\sqrt{2}} \tan t)$, where $t \in (-\frac{\pi}{2}, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \frac{3\pi}{2})$, on the hyperbola E with equation $x^2 - 2y^2 = 1$.

(a) Determine the foci F and F' of E .

(b) Determine the gradients FP and $F'P$, when these lines are not parallel to the y -axis.

(c) Determine the point P in the first quadrant on E for which FP is perpendicular to $F'P$.

Proof. We compare $x^2 - 2y^2 = 1$ with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The vertices occur at $y = 0$, where $x^2 = 1$, so $a^2 = 1$ and $a = 1$. Matching the y -term,

$$2y^2 = \frac{y^2}{b^2} \implies b^2 = \frac{1}{2}.$$

For a hyperbola $b^2 = a^2(e^2 - 1)$, so

$$\frac{1}{2} = 1 \cdot (e^2 - 1) \implies e^2 = \frac{3}{2} \implies e = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}.$$

The foci are at $(\pm ae, 0) = \left(\pm \frac{\sqrt{6}}{2}, 0\right)$. Write

$$F = \left(\frac{\sqrt{6}}{2}, 0\right), \quad F' = \left(-\frac{\sqrt{6}}{2}, 0\right).$$

Proof. With $P = (\sec t, \frac{1}{\sqrt{2}} \tan t)$, the gradient of FP is

$$m_{FP} = \frac{\frac{1}{\sqrt{2}} \tan t - 0}{\sec t - \frac{\sqrt{6}}{2}} = \frac{\frac{1}{\sqrt{2}} \tan t}{\sec t - \frac{\sqrt{6}}{2}},$$

and the gradient of $F'P$ is

$$m_{F'P} = \frac{\frac{1}{\sqrt{2}} \tan t - 0}{\sec t + \frac{\sqrt{6}}{2}} = \frac{\frac{1}{\sqrt{2}} \tan t}{\sec t + \frac{\sqrt{6}}{2}}.$$

Proof. The lines FP and $F'P$ are perpendicular iff the vectors $\vec{FP}$ and $\vec{F'P}$. With $P = (\sec t, \frac{1}{\sqrt{2}} \tan t)$, $F = (\frac{\sqrt{6}}{2}, 0)$, and $F' = (-\frac{\sqrt{6}}{2}, 0)$,

$$\vec{FP} = \left(\sec t - \frac{\sqrt{6}}{2}, \frac{1}{\sqrt{2}} \tan t \right), \quad \vec{F'P} = \left(\sec t + \frac{\sqrt{6}}{2}, \frac{1}{\sqrt{2}} \tan t \right).$$

Their dot product is

$$\vec{FP} \cdot \vec{F'P} = \left(\sec t - \frac{\sqrt{6}}{2} \right) \left(\sec t + \frac{\sqrt{6}}{2} \right) + \frac{1}{2} \tan^2 t = \sec^2 t - \frac{3}{2} + \frac{1}{2} \tan^2 t.$$

Setting this equal to 0 and using $\tan^2 t = \sec^2 t - 1$,

$$\sec^2 t - \frac{3}{2} + \frac{1}{2} (\sec^2 t - 1) = 0 \implies \frac{3}{2} \sec^2 t - 2 = 0 \implies \frac{3}{2} \sec^2 t = 2,$$

so $\sec^2 t = \frac{4}{3}$. In the first quadrant $\sec t > 0$, giving

$$x = \sec t = \frac{2}{\sqrt{3}}, \quad \tan^2 t = \sec^2 t - 1 = \frac{1}{3},$$

so with $\tan t > 0$,

$$y = \frac{1}{\sqrt{2}} \tan t = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{6}}.$$

Thus

$$P = \left(\frac{2}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right).$$

■

1.2 Properties of Conics

Problem 1

Determine the gradient of the tangent to the curve in $\mathbb{R}^2$ with parametric equations

$$x = 2 \cos t + \cos 2t + 1, y = 2 \sin t + \sin 2t$$

at the point with parameter t , where t is not a multiple of π . Hence determine the equation of the tangent to this curve at the points with parameter $t = \pi/3$ and $t = \pi/2$.

Proof. By Theorem 1, the gradient of the tangent is

$$\frac{y'(t)}{x'(t)} = \frac{2 \cos t + 2 \cos(2t)}{-2 \sin t - 2 \sin 2t} = -\frac{\cos t + \cos 2t}{\sin t + \sin 2t}.$$

Then plugging in $t = \pi/3$ we have, since $\cos(\pi/3) + \cos(2\pi/3) = \frac{1}{2} - \frac{1}{2} = 0$, that the gradient is 0, while $x = 3/2$ and $y = 3\sqrt{3}/2$, so the tangent is

$$y = \frac{3\sqrt{3}}{2}.$$

and plugging in $t = \pi/2$ we have, since $\cos(\pi/2) + \cos(\pi) = 0 - 1 = -1$ and $\sin(\pi/2) + \sin(\pi) = 1 + 0 = 1$, that the gradient is 1, while $x = 0$ and $y = 2$, so the tangent is

$$y = x + 2.$$

■

Problem 2

1. Determine the equation of the tangent at a point P on the rectangular parabola with parametric equations $x = t, y = 1/t$.
2. Hence determine the equations of the two tangents to the rectangular hyperbola from the point $(1, -1)$.